

SAME RHYTHMS, LOST COUPLING

Noradrenergic suppression may uncouple pupil size from the infraslow sigma rhythm of human sleep

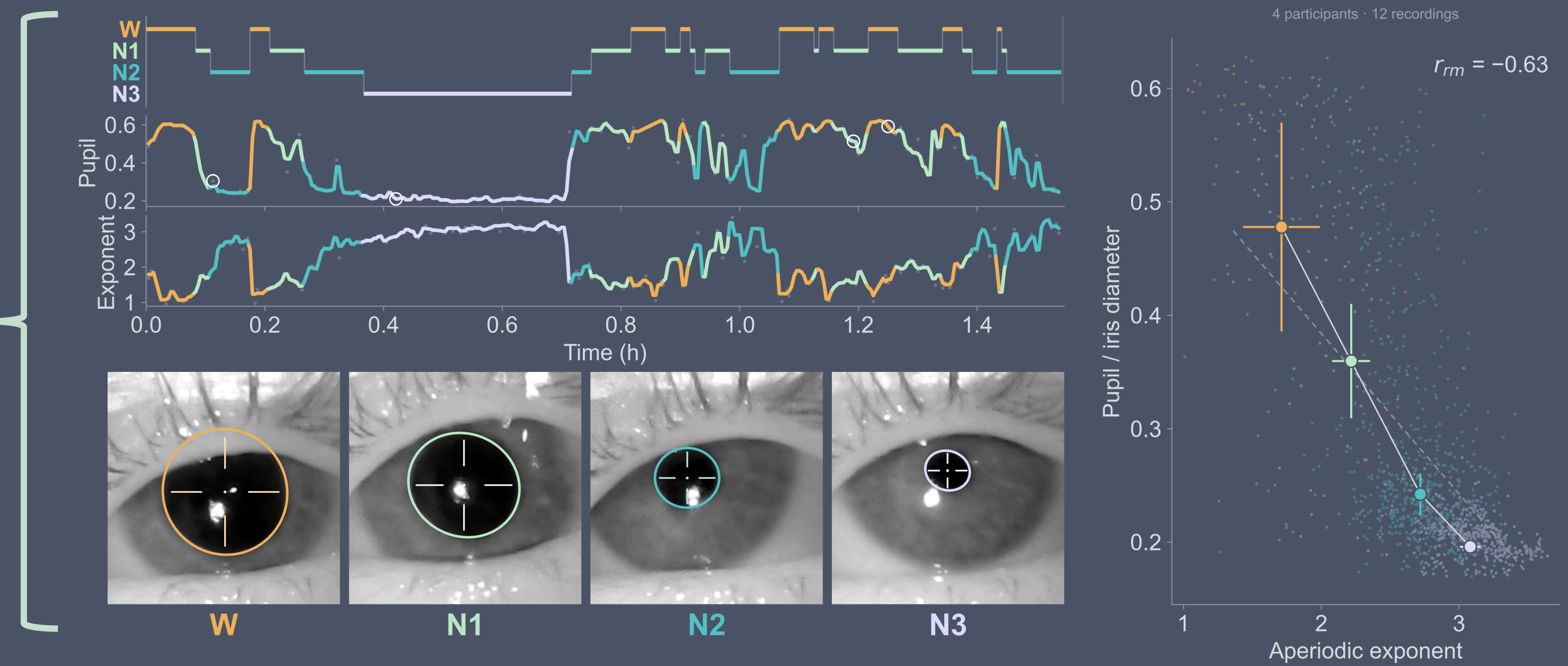
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1 INTRODUCTION

- During sleep, pupil size tracks cortical arousal and has been linked to the locus coeruleus–noradrenergic (LC–NA) system [1].
- LC–NA activity fluctuates on a ~50-s infraslow rhythm that organises micro-arousals and sleep spindles [2–5].

Our pilot replicates the pupil–arousal link: pupil size falls from wake to deep sleep and mirrors the aperiodic exponent, a marker of cortical activity



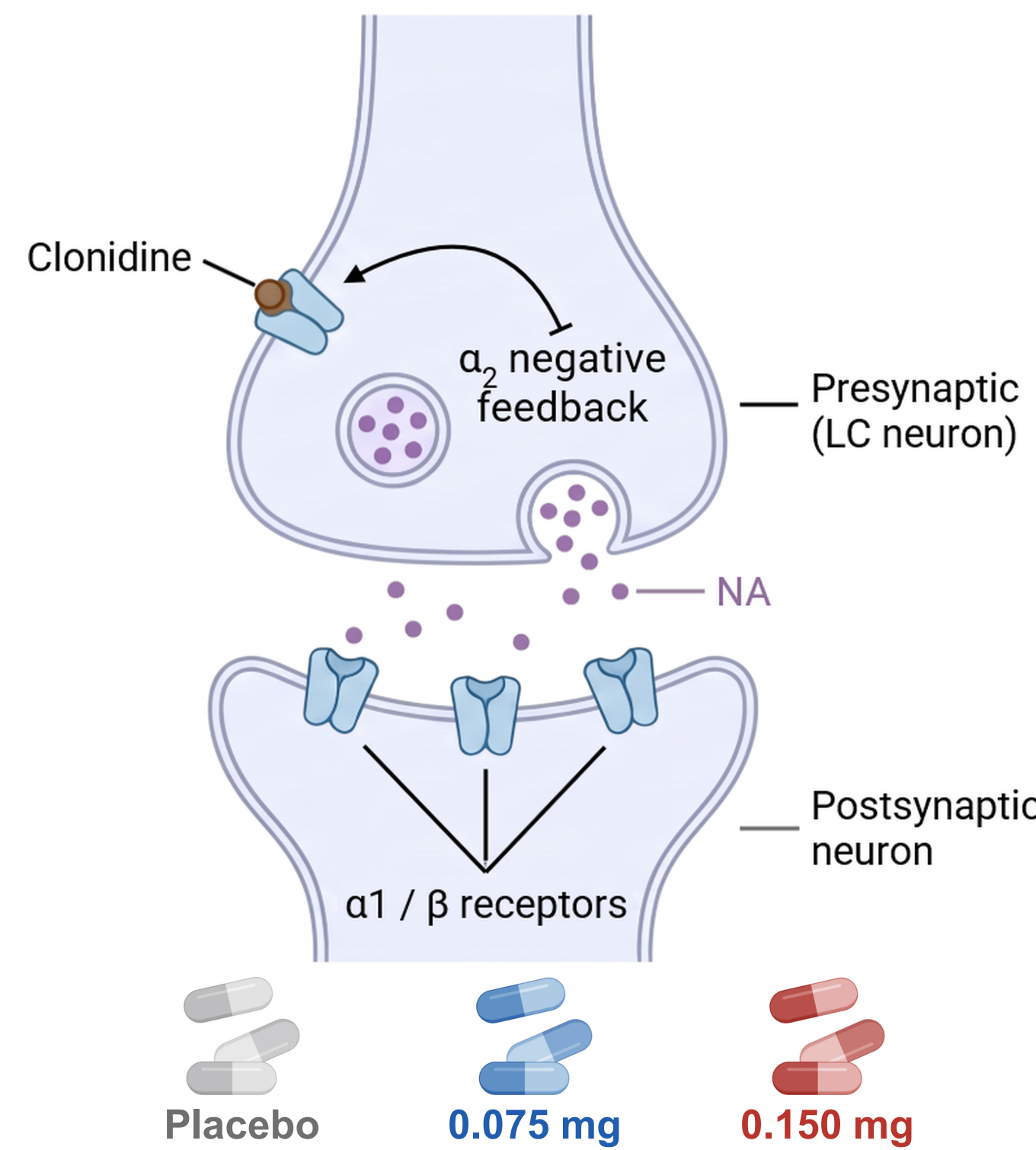
QUESTION

Are pupil changes during sleep causally linked to LC–NA signalling?

We dampened noradrenaline release with clonidine (α_2 -agonist).

Locus coeruleus

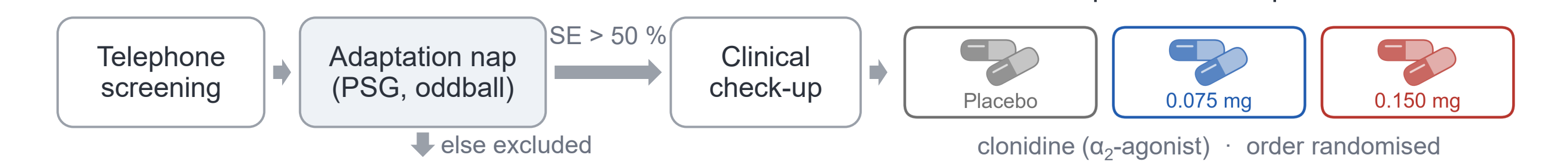
LC–NA SYSTEM & CLONIDINE



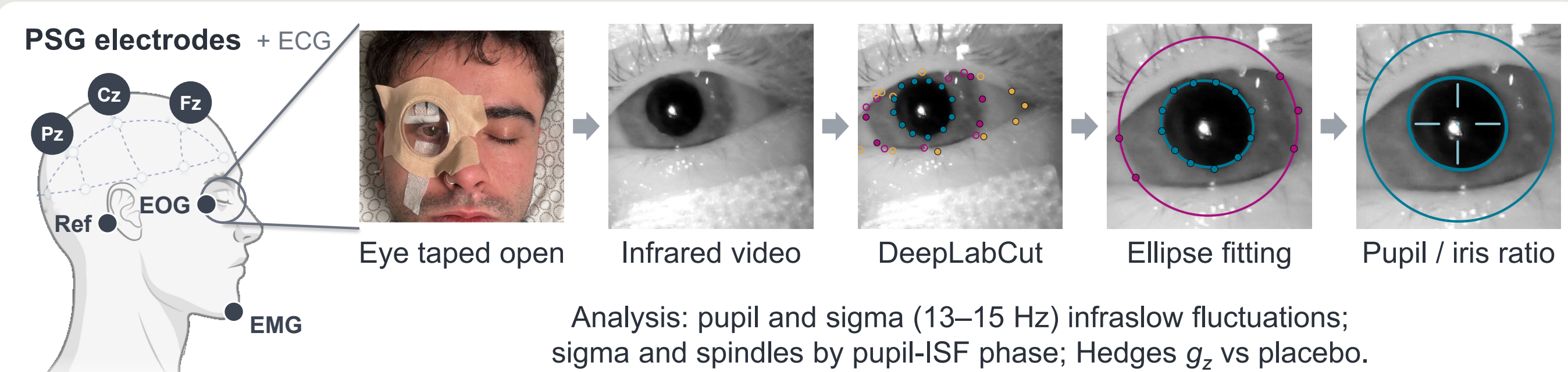
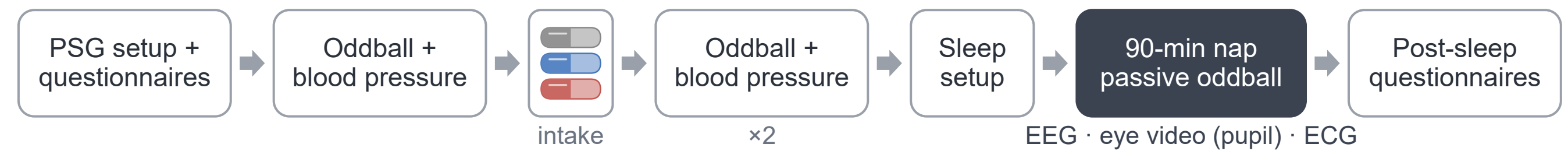
2 METHODS

4 healthy adults · double-blind, placebo-controlled crossover · naps 7.1 ± 1.8 days apart

Within-subject, double-blind, placebo-controlled



Experimental session (x3)

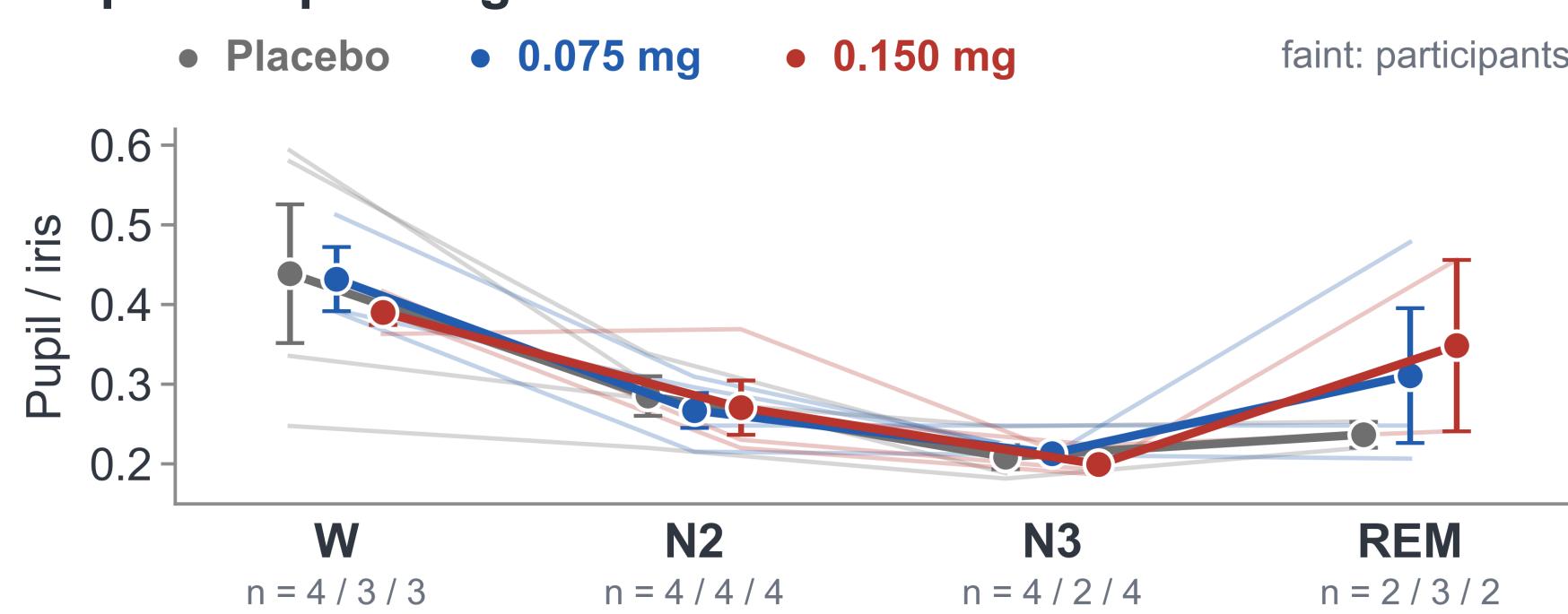


3 RESULT · THE DECOUPLING

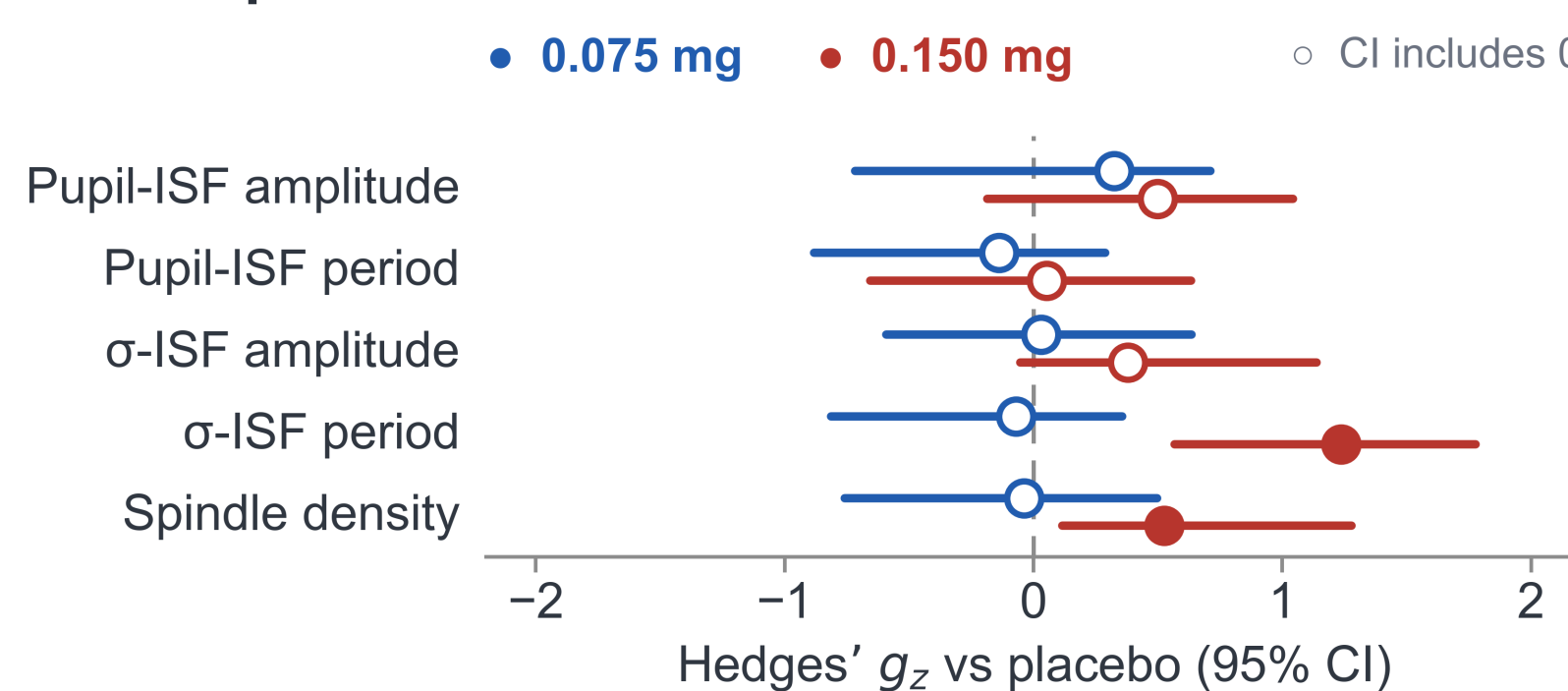
Under 0.150 mg, sigma power and spindles no longer followed the pupil

SAME RHYTHMS pupil size and infraslow rhythms unchanged

a Pupil size per stage

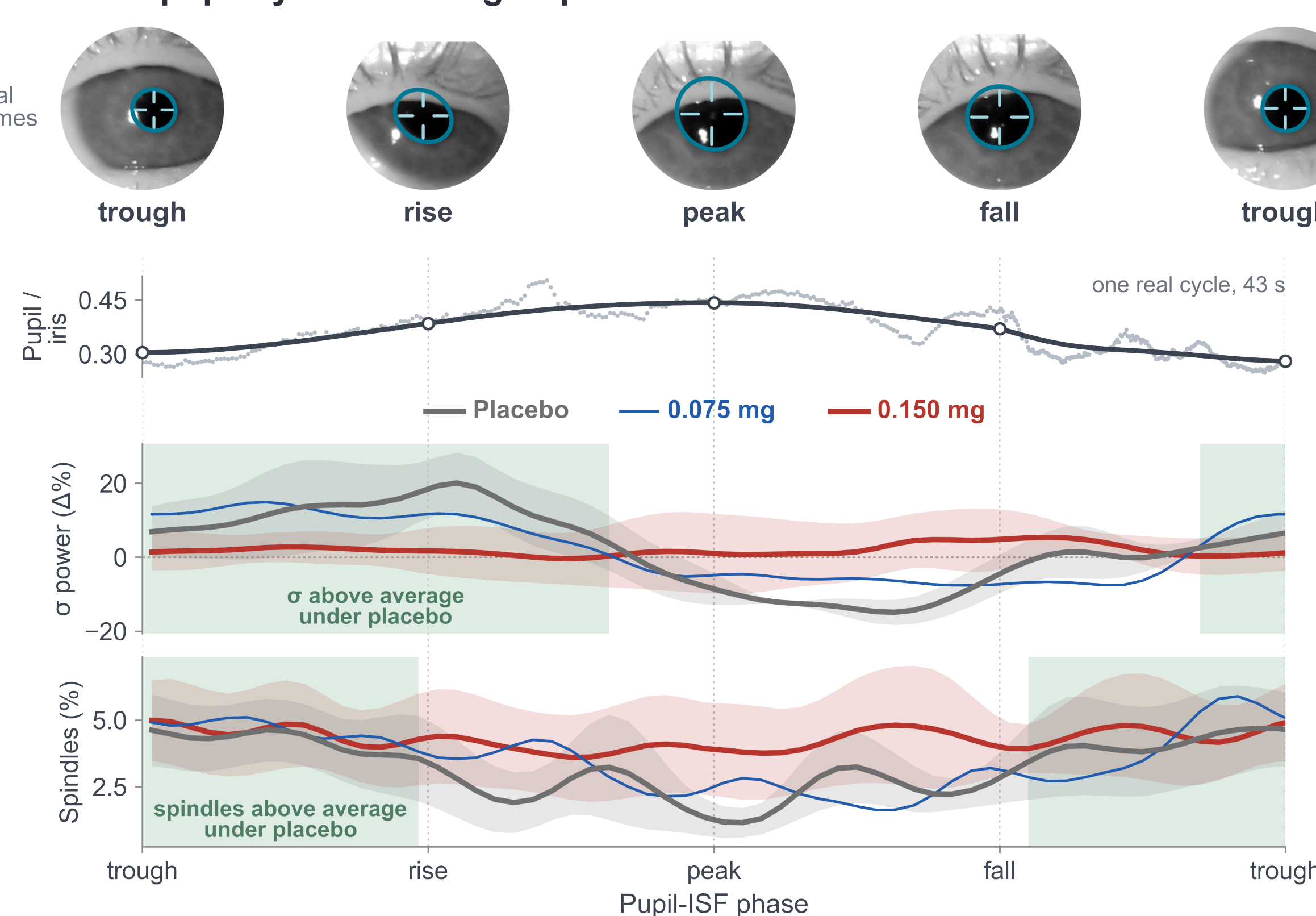


b Rhythms vs placebo

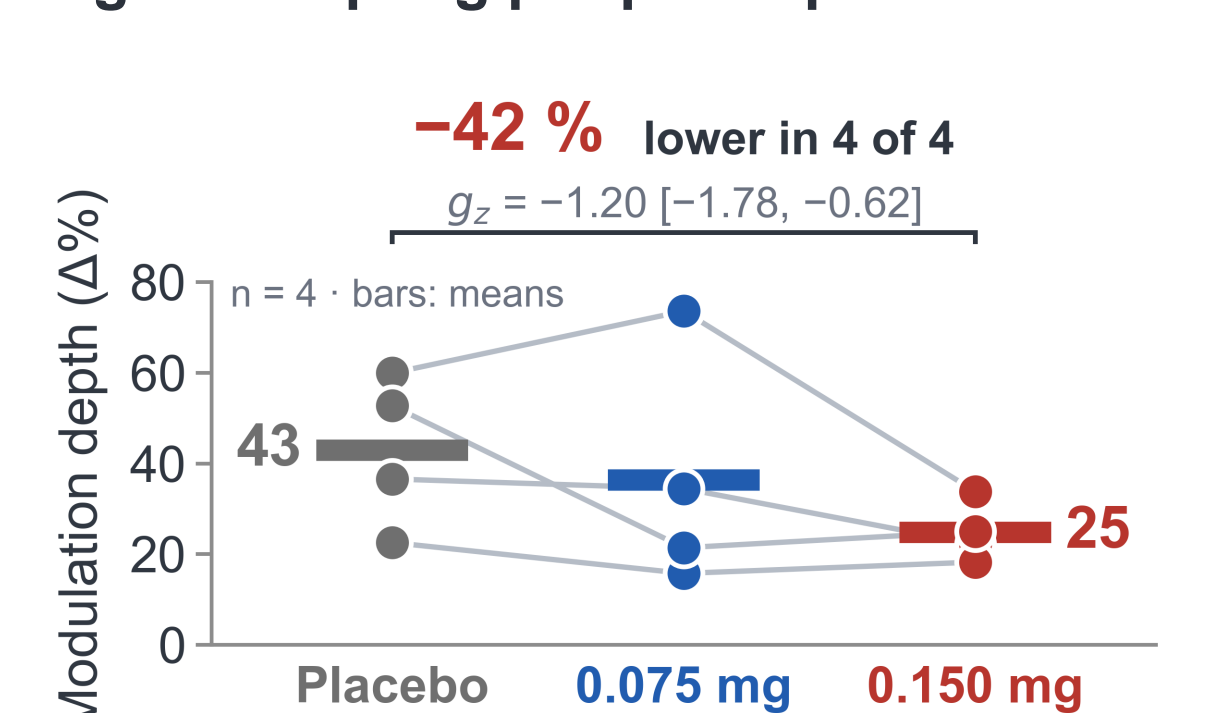


KEY RESULT · LOST COUPLING under 0.150 mg

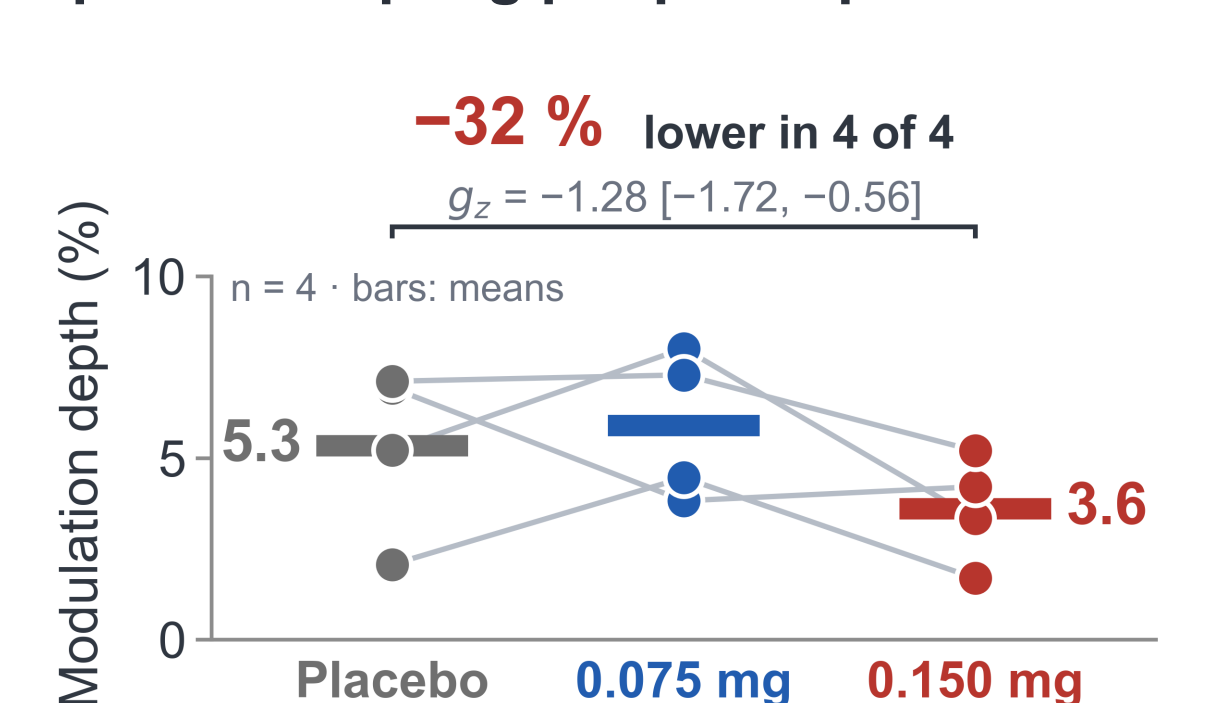
c One real pupil cycle and the group curves



d Sigma coupling per participant

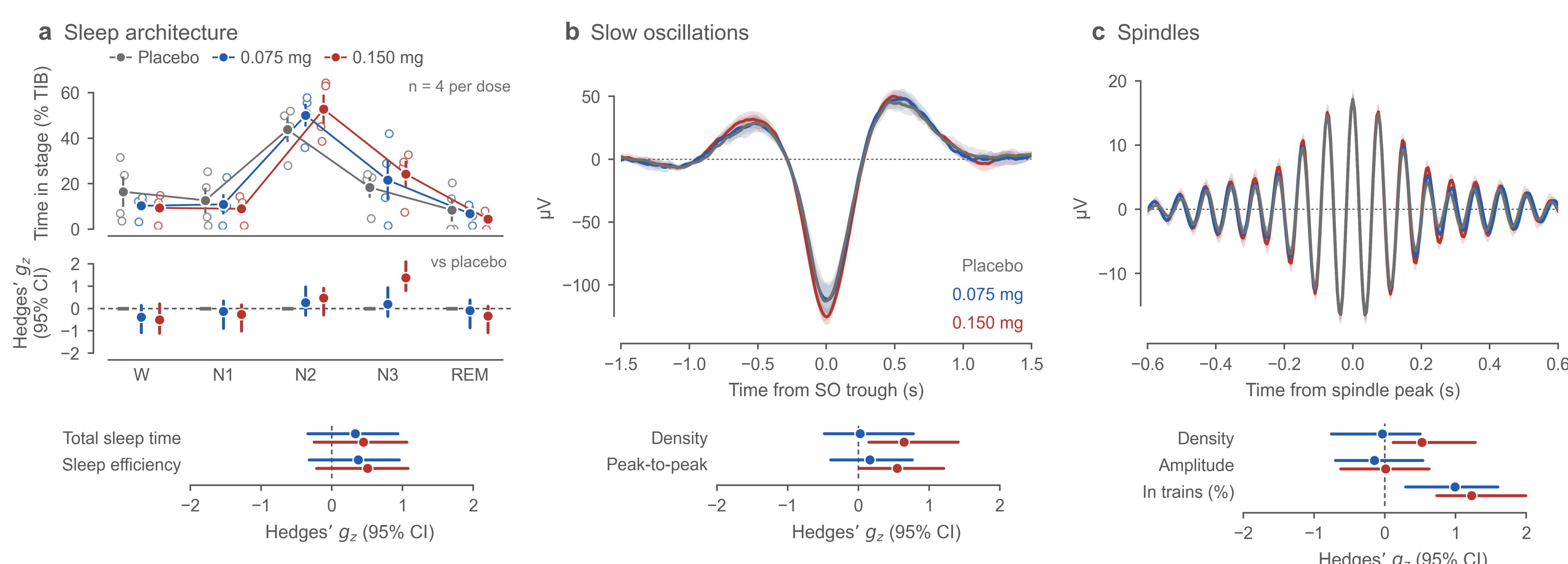


e Spindle coupling per participant



4 SLEEP STAYED LARGELY COMPARABLE

0.150 mg deepened NREM sleep: faster onset, more N3, denser slow oscillations and spindles.



5 TAKEAWAYS

- Pupil size tracks cortical arousal; under placebo, **sigma power followed the pupil's infraslow rhythm**.
- At 0.150 mg the **rhythms and pupil size persisted**, but **sigma power and spindles uncoupled** from the pupil.
- A pupil rhythm can persist without LC–NA drive [7]; **LC–NA may keep neuromodulator rhythms in register** [8].

The coupling, not the oscillations, may depend on noradrenergic signalling.

Next: the ongoing nocturnal study, powered for $g_z = 0.8$ ($n \geq 15$), with a pupil-quality criterion.

Pilot: $n = 4$ daytime naps; pupil tracking varied between naps; clonidine also acts outside the LC.

References

- [1] Carro-Domínguez et al., Nat Commun 2025
- [2] Osorio-Forero et al., Nat Neurosci 2025
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- [7] Yüzgeç et al., Curr Biol 2018
- [8] Kjaerby et al., bioRxiv 2025 (preprint)

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